NAME : MARSYA ZULFA ZAHRANI
NIM : 244107060005
ABSENT : 11
CLASS : SIB 2G
OBJECT : Mobile Programming

Jobsheet Flutter 2
Layout dan Navigasi

Praktikum 1: Membangun Layout di Flutter

Langkah 1: Buat Project Baru

Buatlah sebuah project flutter baru dengan nama layout_flutter. Atau sesuaikan style laporan praktikum yang Anda buat.

Langkah 2: Buka file lib/main.dart

Buka file main.dart lalu ganti dengan kode berikut. Isi nama dan NIM Anda di text title.

```
import 'package:flutter/material.dart';

Run | Debug | Profile
void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter layout: Marsya Zulfa Zahrani 244107060005',
      home: Scaffold(
        appBar: AppBar(
          title: const Text('Flutter layout demo'),
        ), // AppBar
        body: const Center(
          child: Text('Hello World'),
        ), // Center
      ), // Scaffold
    ); // MaterialApp
  }
}
```

Langkah 4: Implementasi title row

Pertama, Anda akan membuat kolom bagian kiri pada judul. Tambahkan kode berikut di bagian atas metode build() di dalam kelas MyApp:

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```
Widget build(BuildContext context) {
  Widget titleSection = Container(
    padding: const EdgeInsets.all(32),
    child: Row(
      children: [
        Expanded(
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Container(
                padding: const EdgeInsets.only(bottom: 8),
                child: const Text(
                  'Wisata Gunung di Batu',
                  style: TextStyle(
                    fontWeight: FontWeight.bold,
                  ), // TextStyle
                ), // Text
              ), // Container
              Text(
                'Batu, Malang, Indonesia',
                style: TextStyle(
                  color: Colors.grey[500],
                ), // TextStyle
              ), // Text
            ],
          ), // Column
        ), // Expanded
        Icon(
          Icons.star,
          color: Colors.red[500],
        ), // Icon
        const Text('41'),
      ],
```

/* soal 1 */ Letakkan widget Column di dalam widget Expanded agar menyesuaikan ruang yang tersisa di dalam widget Row. Tambahkan
properti crossAxisAlignment ke CrossAxisAlignment.start sehingga posisi kolom berada di awal baris.

/* soal 2 */ Letakkan baris pertama teks di dalam Container sehingga memungkinkan Anda untuk menambahkan padding = 8. Teks ‘Batu, Malang, Indonesia' di dalam Column, set warna menjadi abu-abu.

/* soal 3 */ Dua item terakhir di baris judul adalah ikon bintang, set dengan warna merah, dan teks "41". Seluruh baris ada di dalam Container dan beri padding di sepanjang setiap tepinya sebesar 32 piksel. Kemudian ganti isi body text ‘Hello World' dengan
variabel titleSection seperti berikut:

NAME : MARSYA ZULFA ZAHRANI
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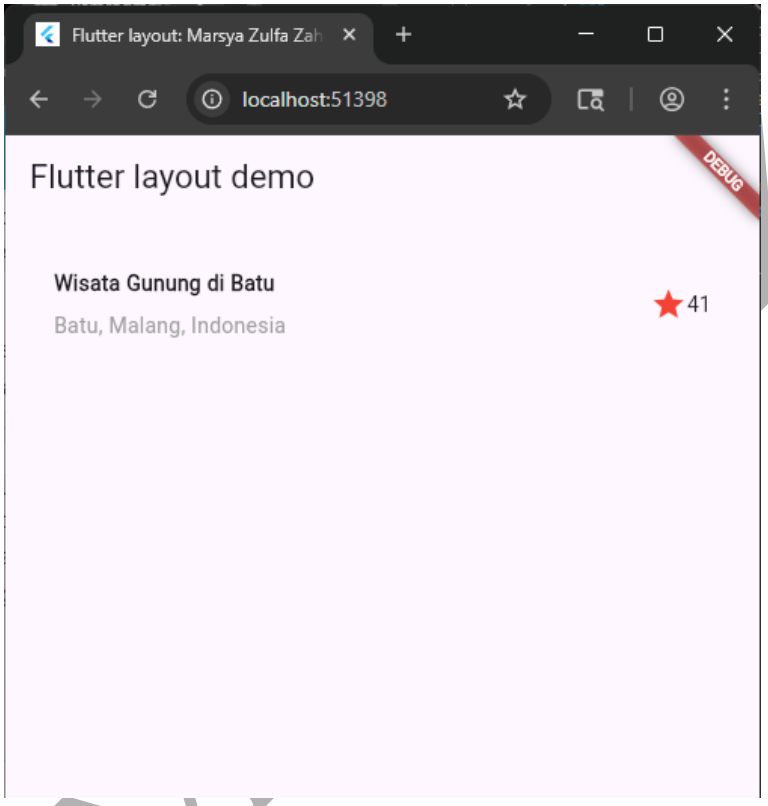
OBJECT : Mobile Programming

```
return MaterialApp(
  title: 'Flutter layout: Marsya Zulfa Zahrani 244107060005',
  home: Scaffold(
    appBar: AppBar(
      title: const Text('Flutter layout demo'),
    ), // AppBar
    body: Column(
      children: [
        titleSection,
      ],
    ), // Column
  ), // Scaffold
); // MaterialApp
}
```



Praktikum 2: Implementasi button row

Langkah 1: Buat method Column _buildButtonColumn

Bagian tombol berisi 3 kolom yang menggunakan tata letak yang sama—sebuah ikon di atas baris teks. Kolom pada baris ini diberi jarak yang sama, dan teks serta ikon diberi warna primer.

Karena kode untuk membangun setiap kolom hampir sama, buatlah metode pembantu pribadi bernama buildButtonColumn(), yang mempunyai parameter warna, Icon dan Text, sehingga dapat mengembalikan kolom dengan widgetnya sesuai dengan warna tertentu.

lib/main.dart (_buildButtonColumn)

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OBJECT : Mobile Programming

```
Column _buildButtonColumn(Color color, IconData icon, String label) {
  return Column(
    mainAxisSize: MainAxisSize.min,
    mainAxisAlignment: MainAxisAlignment.center,
    children: [
      Icon(icon, color: color),
      Container(
        margin: const EdgeInsets.only(top: 8),
        child: Text(
          label,
          style: TextStyle(
            fontSize: 12,
            fontWeight: FontWeight.w400,
            color: color,
          ), // TextStyle
        ), // Text
      ), // Container
    ],
  ); // Column
}
```

Langkah 2: Buat widget buttonSection

Buat Fungsi untuk menambahkan ikon langsung ke kolom. Teks berada di dalam Container dengan margin hanya di bagian atas, yang memisahkan teks dari ikon.

Bangun baris yang berisi kolom-kolom ini dengan memanggil fungsi dan set warna, Icon, dan teks khusus melalui parameter ke kolom tersebut. Sejajarkan kolom di sepanjang sumbu utama menggunakan MainAxisAlignment.spaceEvenly untuk mengatur ruang kosong secara merata sebelum, di antara, dan setelah setiap kolom. Tambahkan kode berikut tepat di bawah deklarasi titleSection di dalam metode build():

lib/main.dart (buttonSection)

```
Color color = Theme.of(context).primaryColor;

Widget buttonSection = Row(
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,
  children: [
    _buildButtonColumn(color, Icons.call, 'CALL'),
    _buildButtonColumn(color, Icons.near_me, 'ROUTE'),
    _buildButtonColumn(color, Icons.share, 'SHARE'),
  ],
);
```

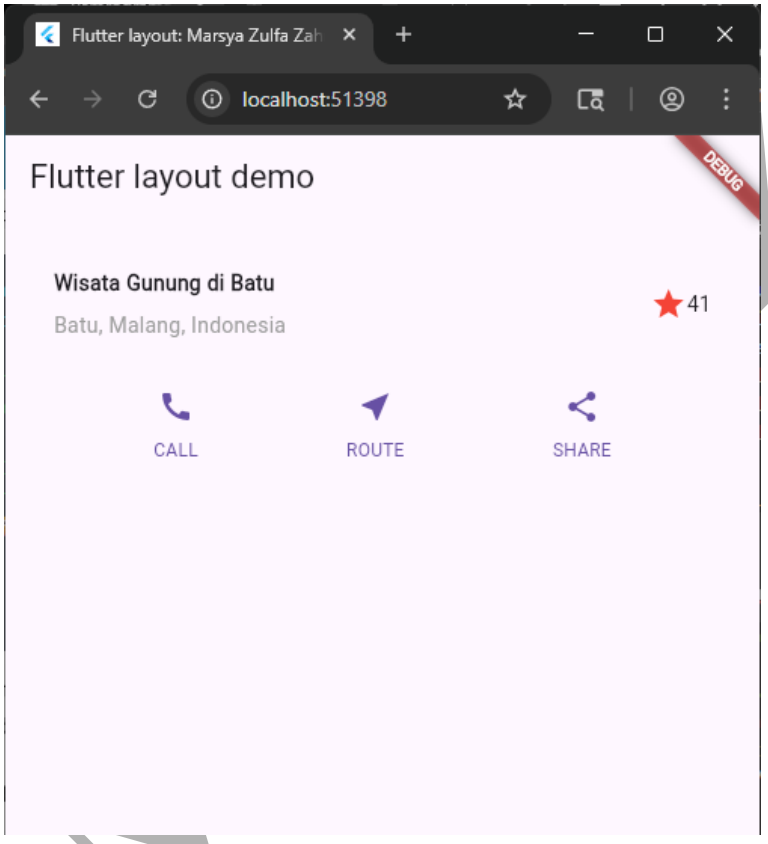
Langkah 3: Tambah button section ke body

Tambahkan variabel buttonSection ke dalam body seperti berikut:

rsya Zulfa

NAME : MARSYA ZULFA ZAHRANI
NIM : 244107060005
ABSENT : 11
CLASS : SIB 2G
OBJECT : Mobile Programming

```
return MaterialApp(
  title: 'Flutter layout: Marsya Zulfa Zahrani 244107060005',
  home: Scaffold(
    appBar: AppBar(
      title: const Text('Flutter layout demo'),
    ), // AppBar
    body: Column(
      children: [
        titleSection,
        buttonSection,
      ],
    ), // Column
  ), // Scaffold
); // MaterialApp
```



Praktikum 3: Implementasi text section

Langkah 1: Buat widget textSection

Tentukan bagian teks sebagai variabel. Masukkan teks ke dalam Container dan tambahkan padding di sepanjang setiap tepinya. Tambahkan kode berikut tepat di bawah deklarasi buttonSection:

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ABSENT : 11
CLASS : SIB 2G
OBJECT : Mobile Programming

```
Widget textSection = Container(
  padding: const EdgeInsets.all(32),
  child: const Text(
    'Easily accessible waterfall flowing '
    'over mossy rocks into a turquoise '
    'basin, with viewing platforms. '
    'Marsya Zulfa Zahrani 244107060005',
    softWrap: true,
  ), // Text
); // Container
```

Dengan memberi nilai softWrap = true, baris teks akan memenuhi lebar kolom sebelum membungkusnya pada batas kata.

Langkah 2: Tambahkan variabel text section ke body

Tambahkan widget variabel textSection ke dalam body seperti berikut:

```
return MaterialApp(
  title: 'Flutter layout: Marsya Zulfa Zahrani 244107060005',
  home: Scaffold(
    appBar: AppBar(
      title: const Text('Flutter layout demo'),
    ), // AppBar
    body: Column(
      children: [
        titleSection,
        buttonSection,
        textSection,
      ],
    ), // Column
  ), // Scaffold
); // MaterialApp
```

Marsya Zulfa

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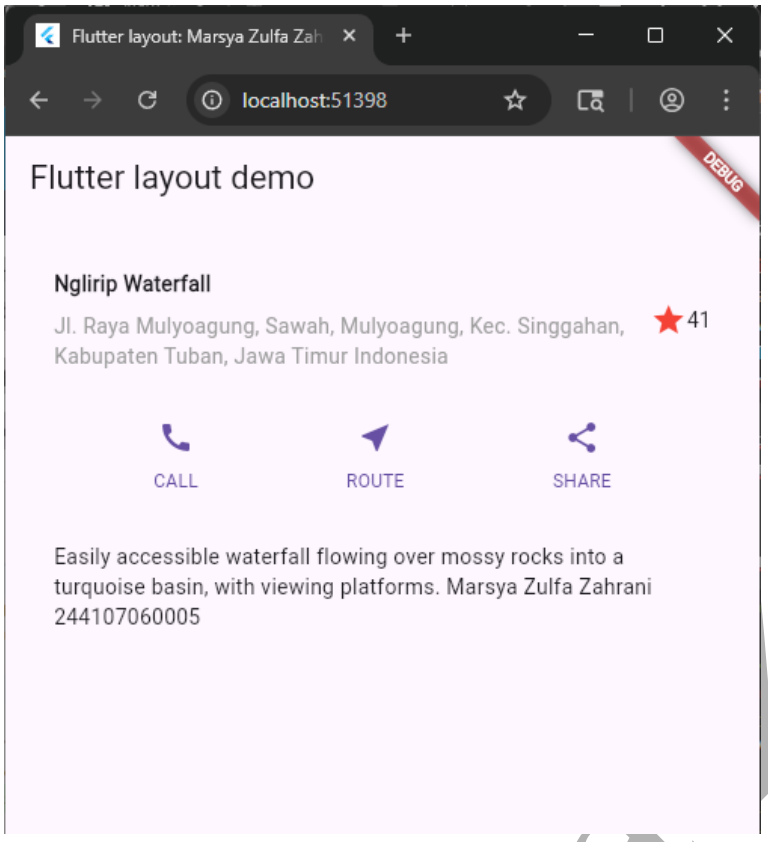

Praktikum 4: Implementasi image section

Langkah 1: Siapkan aset gambar

Anda dapat mencari gambar di internet yang ingin ditampilkan. Buatlah folder images di root project layout_flutter. Masukkan file gambar tersebut ke folder images, lalu set nama file tersebut ke file pubspec.yaml seperti berikut:

```
flutter:
  uses-material-design: true

  assets:
    - images/waterfall.jpg
```

Langkah 2: Tambahkan gambar ke body

Tambahkan aset gambar ke dalam body seperti berikut:

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CLASS : SIB 2G
OBJECT : Mobile Programming

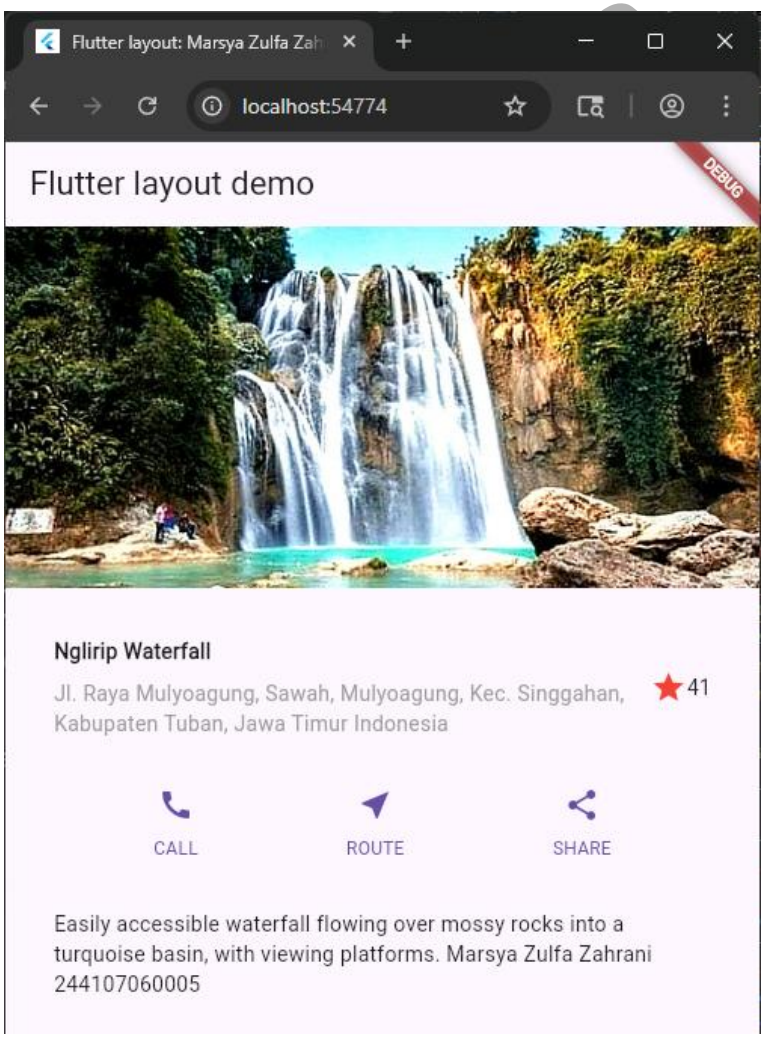
```
body: ListView(
  children: [
    Image.asset(
      'images/waterfall.jpg',
      width: 600,
      height: 240,
      fit: BoxFit.cover,
      ), // Image.asset
    titleSection,
    buttonSection,
    textSection,
```

BoxFit.cover memberi tahu kerangka kerja bahwa gambar harus sekecil mungkin tetapi menutupi seluruh kotak rendernya.

Langkah 3: Terakhir, ubah menjadi ListView

Pada langkah terakhir ini, atur semua elemen dalam ListView, bukan Column, karena ListView mendukung scroll yang dinamis saat aplikasi dijalankan pada perangkat yang resolusinya lebih kecil.

```
body: ListView(
```



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Tugas Praktikum 1

Implementasikan di project baru "basic_layout_flutter" dengan mengakses sumber ini: https://docs.flutter.dev/codelabs/layout-basics

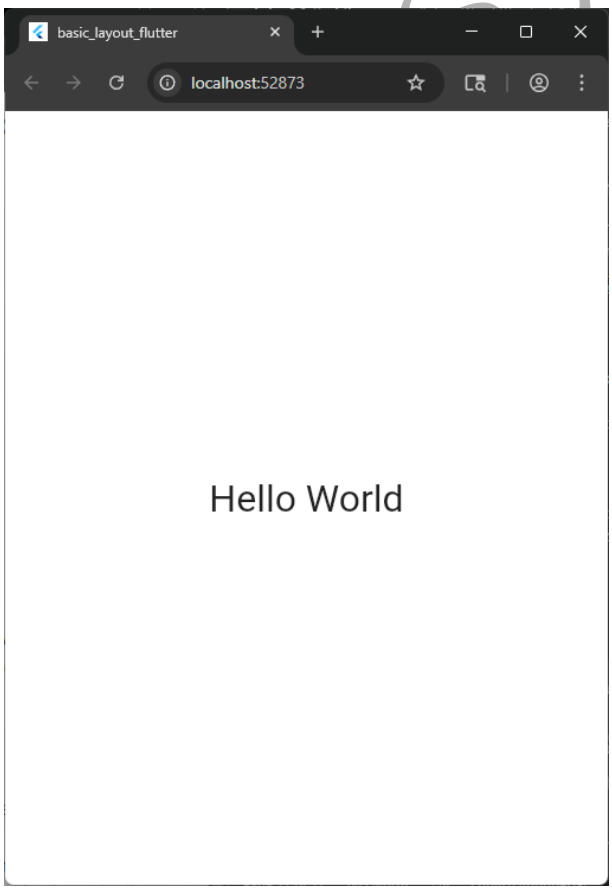
Lay out a widget

A Flutter app is itself a widget, and most widgets have a build() method. Instantiating and returning a widget in the app's build() method displays the widget.

```
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return Container(
      decoration: const BoxDecoration(color: Colors.white),
      child: const Center(
        child: Text(
          'Hello World',
          textDirection: TextDirection.ltr,
          style: TextStyle(fontSize: 32, color: Colors.black87),
        ), // Text
      ), // Center
    ); // Container
  }
}
```

After you've added your widgets, run your app. When you run the app, you should see Hello World.



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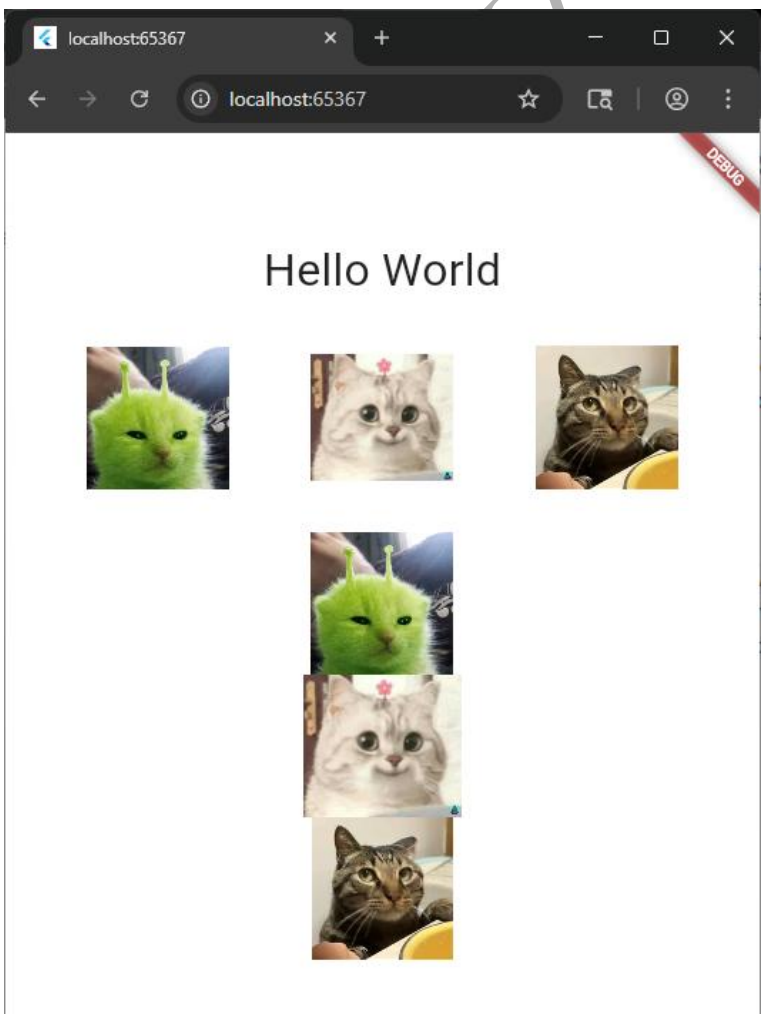
Lay out multiple widgets vertically and horizontally

One of the most common layout patterns is to arrange widgets vertically or horizontally. You can use a Row widget to arrange widgets horizontally, and a Column widget to arrange widgets vertically.

```
const SizedBox(height: 30),

Row(
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,
  children: [
    Image.asset('images/pict1.jpg', width: 100),
    Image.asset('images/pict2.jpg', width: 100),
    Image.asset('images/pict3.png', width: 100),
  ],
), // Row
const SizedBox(height: 30),

Column(
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,
  children: [
    Image.asset('images/pict1.jpg', height: 100),
    Image.asset('images/pict2.jpg', height: 100),
    Image.asset('images/pict3.png', height: 100),
  ],
), // Column
```



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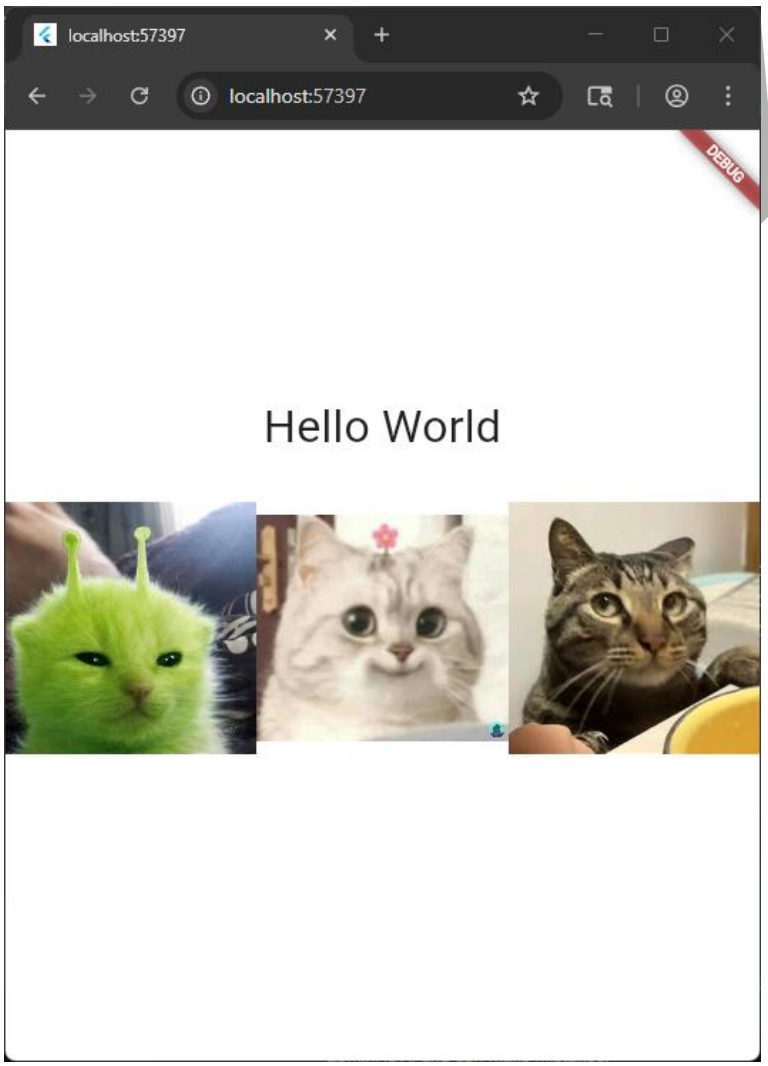
Sizing widgets

When a layout is too large to fit a device, a yellow and black striped pattern appears along the affected edge. Widgets can be sized to fit within a row or column by using the Expanded widget. To fix the previous where the row of images is too wide for its render box, wrap each image with an Expanded widget.

```
Row(
  crossAxisAlignment: CrossAxisAlignment.center,
  children: [
    Expanded(child: Image.asset('images/pict1.jpg')),
    Expanded(child: Image.asset('images/pict2.jpg')),
    Expanded(child: Image.asset('images/pict3.png')),
  ],
), // Row
```



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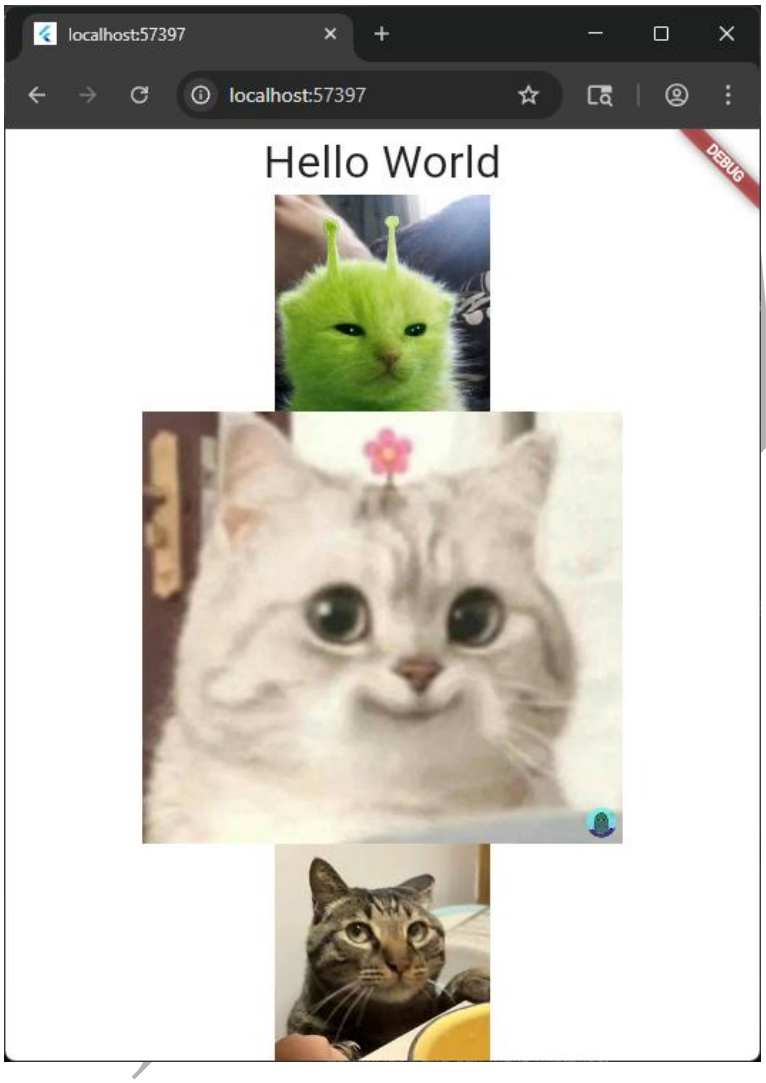
```
Expanded(
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.center,
    children: [
      Expanded(child: Image.asset('images/pict1.jpg')),
      Expanded(flex: 2, child: Image.asset('images/pict2.jpg')),
      Expanded(child: Image.asset('images/pict3.png')),
    ],
  ), // Column
), // Expanded
```



Zulfa

Packing widgets

By default, a row or column occupies as much space along its main axis as possible, but if you want to pack the children closely together, set its mainAxisSize to MainAxisSize.min. The following example uses this property to pack the star icons together.

NAME : MARSYA ZULFA ZAHRANI
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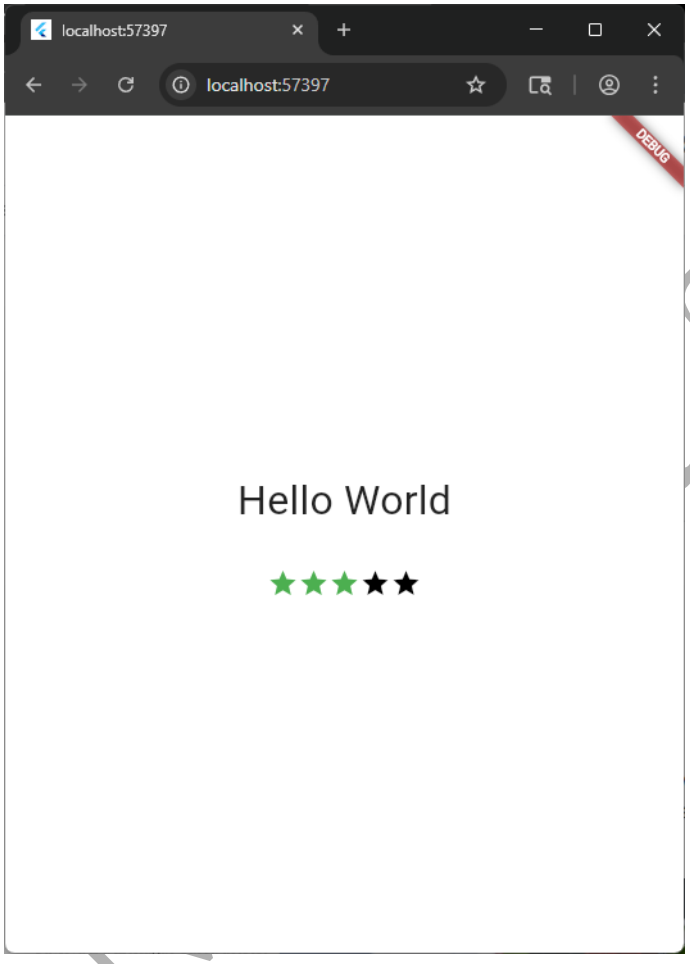
OBJECT : Mobile Programming

```
Row(
  mainAxisSize: MainAxisSize.min,
  children: [
    Icon(Icons.star, color: Colors.green[500]),
    Icon(Icons.star, color: Colors.green[500]),
    Icon(Icons.star, color: Colors.green[500]),
    const Icon(Icons.star, color: Colors.black),
    const Icon(Icons.star, color: Colors.black),
  ],
), // Row
```



Nesting rows and columns

The layout framework allows you to nest rows and columns inside of rows and columns as deeply as you need.

The outlined section is implemented as two rows. The ratings row contains five stars and the number of reviews. The icons row contains three columns of icons and text.

The widget tree for the ratings row:

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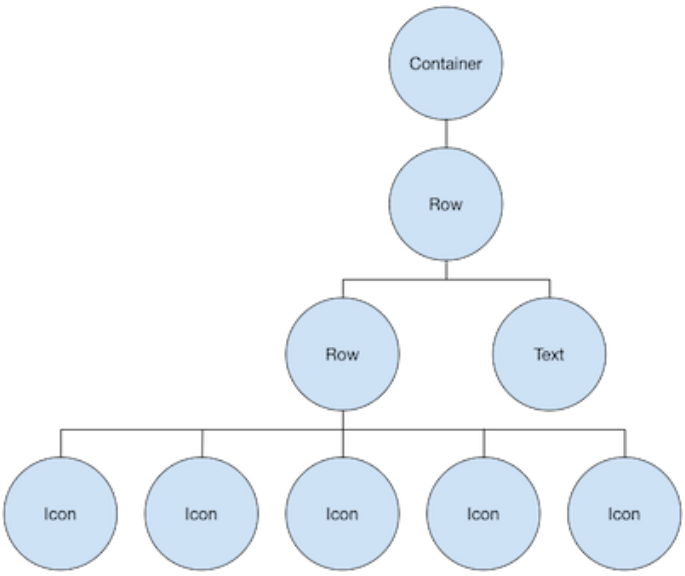

The ratings variable creates a row containing a smaller row of 5-star icons, and text:

```
final stars = Row(
  mainAxisSize: MainAxisSize.min,
  children: [
    Icon(Icons.star, color: Colors.green[500]),
    Icon(Icons.star, color: Colors.green[500]),
    Icon(Icons.star, color: Colors.green[500]),
    const Icon(Icons.star, color: Colors.black),
    const Icon(Icons.star, color: Colors.black),
  ],
); // Row

final ratings = Container(
  padding: const EdgeInsets.all(20),
  child: Row(
    mainAxisAlignment: MainAxisAlignment.spaceEvenly,
    children: [
      stars,
      const Text(
        '170 Reviews',
        style: TextStyle(
          color: Colors.black,
          fontWeight: FontWeight.w800,
          fontSize: 20,
        ), // TextStyle
      ), // Text
    ],
  ), // Row
); // Container
```

The icons row, below the ratings row, contains 3 columns; each column contains an icon and two lines of text, as you can see in its widget tree:

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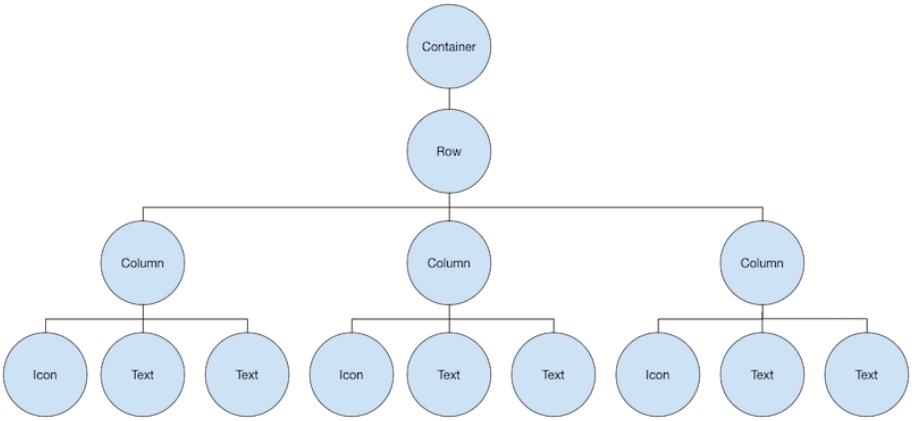

The iconList variable defines the icons row:

```
const descTextStyle = TextStyle(
  color: Colors.black,
  fontWeight: FontWeight.w800,
  fontFamily: 'Roboto',
  letterSpacing: 0.5,
  fontSize: 18,
  height: 2,
);

// DefaultTextStyle.merge() allows you to create a default text
// style that is inherited by its child and all subsequent children.
```

```
final iconList = DefaultTextStyle.merge(
  style: descTextStyle,
  child: Container(
    padding: const EdgeInsets.all(20),
    child: Row(
      mainAxisAlignment: MainAxisAlignment.spaceEvenly,
      children: [
        Column(
          children: [
            Icon(Icons.kitchen, color: Colors.green[500]),
            const Text('PREP:'),
            const Text('25 min'),
          ],
        ), // Column
        Column(
          children: [
            Icon(Icons.timer, color: Colors.green[500]),
            const Text('COOK:'),
            const Text('1 hr'),
          ],
        ), // Column
        Column(
          children: [
            Icon(Icons.restaurant, color: Colors.green[500]),
            const Text('FEEDS:'),
            const Text('4-6'),
          ],
        ), // Column
      ],
    ), // Row
  ), // Container
);
```

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The leftColumn variable contains the ratings and icons rows, as well as the title and text that describes the Pavlova:

```
final leftColumn = Container(
  padding: const EdgeInsets.fromLTRB(20, 30, 20, 20),
  child: Column(
    children: [titleText, subTitle, ratings, iconList],
  ), // Column
); // Container
```

The left column is placed in a SizedBox to constrain its width. Finally, the UI is constructed with the entire row (containing the left column and the image) inside a Card.

The Pavlova image is from Pixabay. You can embed an image from the net using Image.network() but, for this example, the image is saved to an images directory in the project, added to the pubspec file, and accessed using Images.asset().

```
body: Center(
  child: Container(
    margin: const EdgeInsets.fromLTRB(0, 40, 0, 30),
    height: 600,
    child: Card(
      child: Row(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          SizedBox(width: 440, child: leftColumn),
          mainImage,
        ],
      ), // Row
    ), // Card
  ), // Container
), // Center
```

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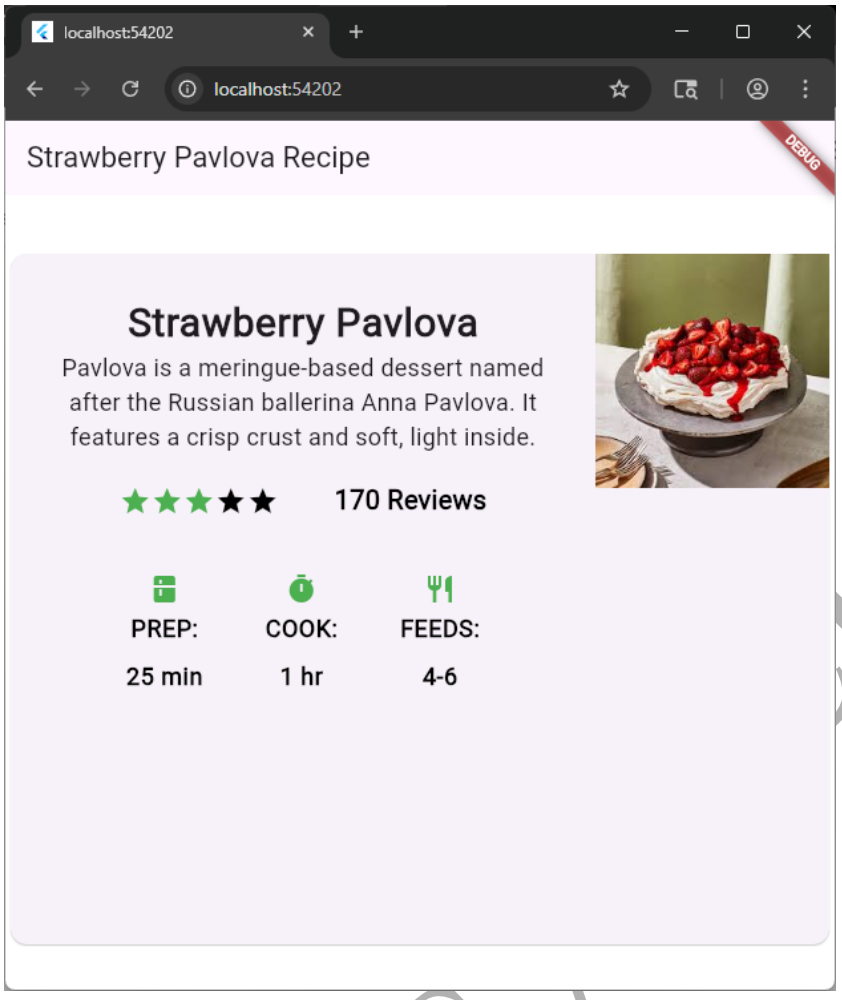

Common layout widgets

The widgets fall into two categories: standard widgets from the widgets library, and specialized widgets from the Material library. Any app can use the widgets library but only Material apps can use the Material Components library.

Container

Many layouts make liberal use of Containers to separate widgets using padding, or to add borders or margins. You can change the device's background by placing the entire layout into a Container and changing its background color or image.

Summary (Container)

- Add padding, margins, borders
- Change background color or image
- Contains a single child widget, but that child can be a Row, Column, or even the root of a widget tree

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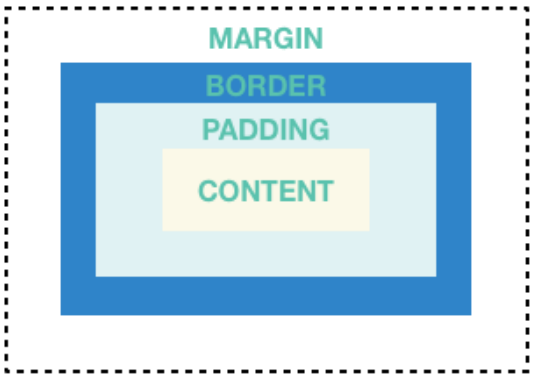
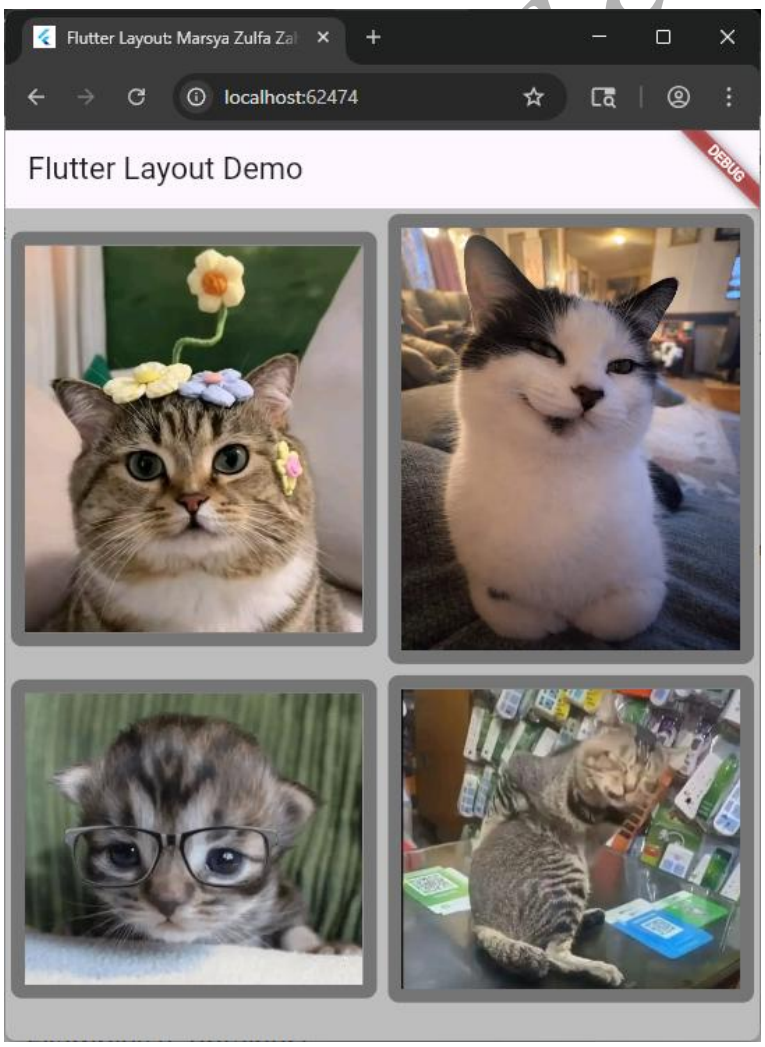


Examples (Container)

This layout consists of a column with two rows, each containing 2 images. A Container is used to change the background color of the column to a lighter grey.

```
Widget _buildImageColumn() => Container(
    decoration: const BoxDecoration(color: Colors.black26),
    child: Column(
      children: [
        _buildImageRow(1),
        _buildImageRow(3),
      ],
    ), // Column
  ); // Container
```



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A Container is also used to add a rounded border and margins to each image:

```
Widget _buildDecoratedImage(int imageIndex) => Expanded(
    child: Container(
      decoration: BoxDecoration(
        border: Border.all(width: 10, color: Colors.black38),
        borderRadius: const BorderRadius.all(Radius.circular(8)),
      ), // BoxDecoration
      margin: const EdgeInsets.all(4),
      child: Image.asset('images/pict$imageIndex.jpeg', fit: BoxFit.cover),
    ), // Container
  ); // Expanded

Widget _buildImageRow(int imageIndex) => Row(
    children: [
      _buildDecoratedImage(imageIndex),
      _buildDecoratedImage(imageIndex + 1),
    ],
  );
```

GridView

Use GridView to lay widgets out as a two-dimensional list. GridView provides two pre-fabricated lists, or you can build your own custom grid. When a GridView detects that its contents are too long to fit the render box, it automatically scrolls.

Summary (GridView)

- Lays widgets out in a grid
- Detects when the column content exceeds the render box and automatically provides scrolling
- Build your own custom grid, or use one of the provided grids:
    - GridView.count allows you to specify the number of columns
    - GridView.extent allows you to specify the maximum pixel width of a tile

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ABSENT : 11
CLASS : SIB 2G
OBJECT : Mobile Programming

```dart
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  Widget _buildGrid() => GridView.extent(
      maxCrossAxisExtent: 150,
      padding: const EdgeInsets.all(4),
      mainAxisSpacing: 4,
      crossAxisSpacing: 4,
      children: _buildGridTileList(30),
    ); // GridView.extent

  List<Widget> _buildGridTileList(int count) => List.generate(
      count,
      (i) => Image.asset(
        'images/pict$i.jpeg',
        fit: BoxFit.cover,
      ), // Image.asset
    ); // List.generate

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Layout: Marsya Zulfa Zahrani',
      home: Scaffold(
        appBar: AppBar(
          title: const Text('GridView Demo'),
        ), // AppBar
        backgroundColor: Colors.white,
        body: _buildGrid(),
      ), // Scaffold
    ); // MaterialApp
  }
}
```

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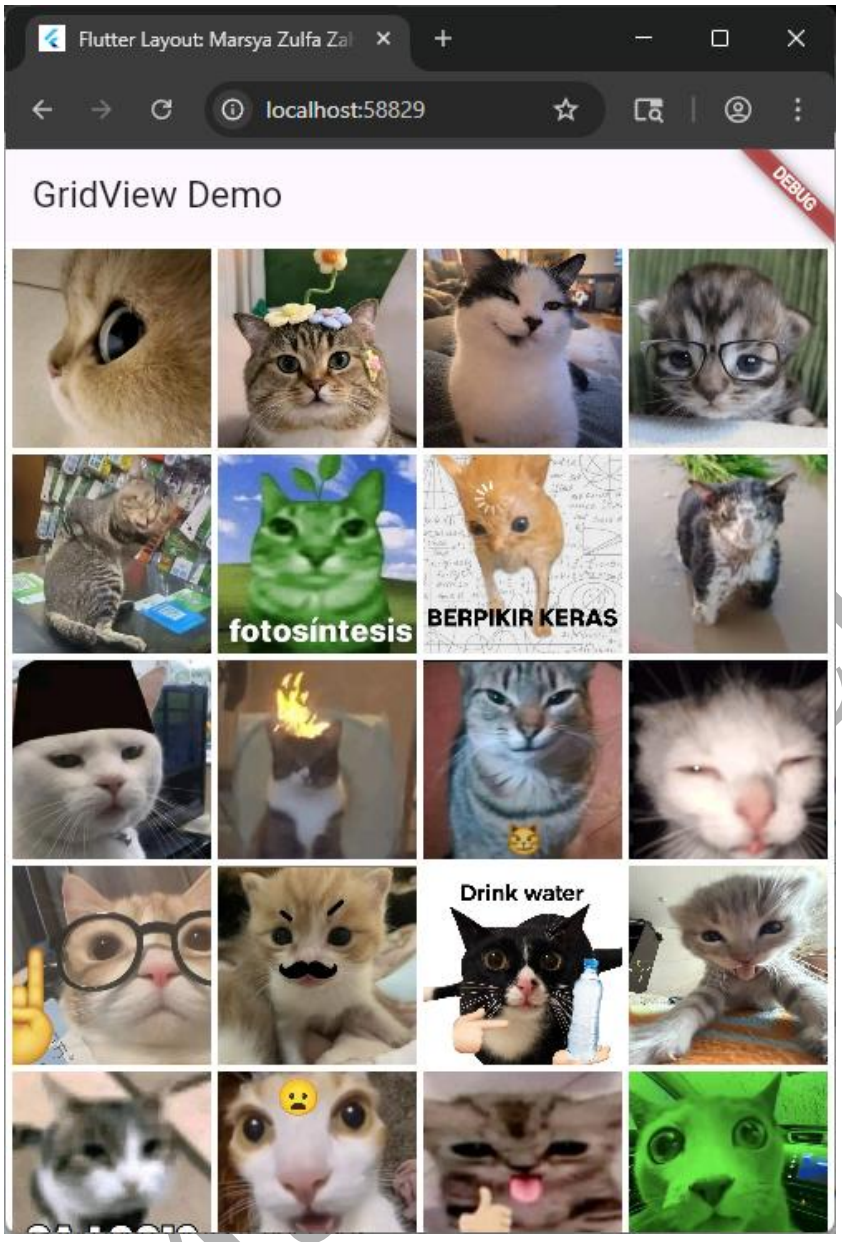

ListView

ListView, a column-like widget, automatically provides scrolling when its content is too long for its render box.

Summary (ListView)

- A specialized Column for organizing a list of boxes
- Can be laid out horizontally or vertically
- Detects when its content won't fit and provides scrolling
- Less configurable than Column, but easier to use and supports scrolling

Uses ListView to display a list of businesses using ListTiles. A Divider separates the theaters from the restaurants.

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OBJECT : Mobile Programming

colors_demo.dart > ...

```dart
import 'package:flutter/material.dart';

class ColorsDemo extends StatelessWidget {
  const ColorsDemo({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('ListView Demo'),
      ), // AppBar
      body: _buildList(),
    ); // Scaffold
  }

  Widget _buildList() {
    return ListView(
      children: [
        _tile('CineArts at the Empire', '85 W Portal Ave', Icons.theaters),
        _tile('The Castro Theater', '429 Castro St', Icons.theaters),
        _tile('Alamo Drafthouse Cinema', '2550 Mission St', Icons.theaters),
        _tile('Roxie Theater', '3117 16th St', Icons.theaters),
        _tile('United Artists Stonestown Twin', '501 Buckingham Way', Icons.theaters),
        _tile('AMC Metreon 16', '135 4th St #3000', Icons.theaters),
        const Divider(),
        _tile('K\'s Kitchen', '757 Monterey Blvd', Icons.restaurant),
        _tile('Emmy\'s Restaurant', '1923 Ocean Ave', Icons.restaurant),
        _tile('Chaiya Thai Restaurant', '272 Claremont Blvd', Icons.restaurant),
        _tile('La Ciccia', '291 30th St', Icons.restaurant),
      ],
    ); // ListView
  }
```

```dart
  ListTile _tile(String title, String subtitle, IconData icon) {
    return ListTile(
      title: Text(
        title,
        style: const TextStyle(fontWeight: FontWeight.w500, fontSize: 20),
      ), // Text
      subtitle: Text(subtitle),
      leading: Icon(icon, color: Colors.blue[500]),
    ); // ListTile
  }
}
```

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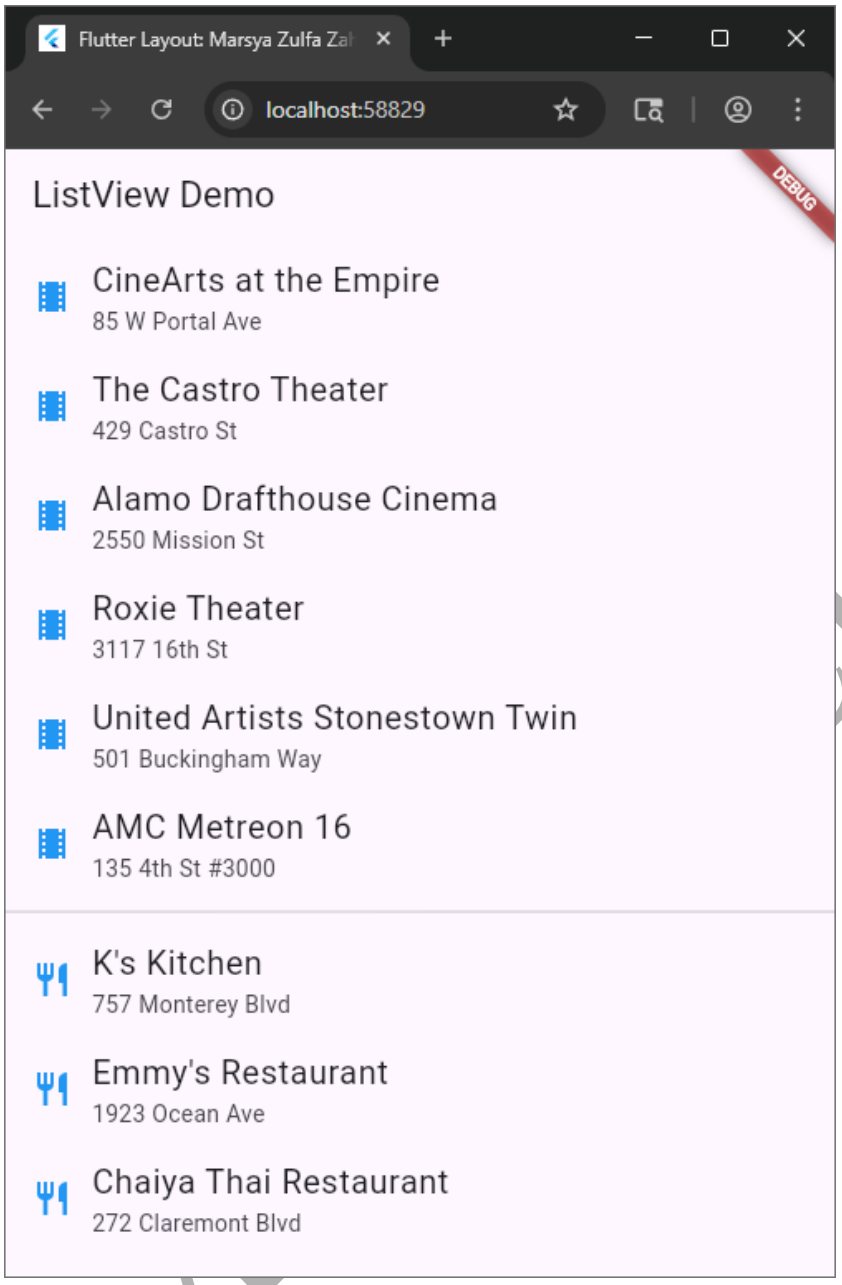
CLASS : SIB 2G
OBJECT : Mobile Programming



Stack

Use Stack to arrange widgets on top of a base widget—often an image. The widgets can completely or partially overlap the base widget.

Summary (Stack)

- Use for widgets that overlap another widget
- The first widget in the list of children is the base widget; subsequent children are overlaid on top of that base widget
- A Stack's content can't scroll
- You can choose to clip children that exceed the render box

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CLASS : SIB 2G
OBJECT : Mobile Programming

bottom_navigation_demo.dart > BottomNavigationDemo

```dart
import 'package:flutter/material.dart';

class BottomNavigationDemo extends StatelessWidget {
  const BottomNavigationDemo({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Stack Widget Demo'),
      ), // AppBar
      body: Center(
        child: _buildStack(),
      ), // Center
    ); // Scaffold
  }
```

```dart
  Widget _buildStack() {
    return Stack(
      alignment: const Alignment(0.6, 0.6),
      children: [
        const CircleAvatar(
          backgroundImage: AssetImage('images/pict9.jpeg'),
          radius: 100,
        ), // CircleAvatar
        Container(
          padding: const EdgeInsets.symmetric(horizontal: 8, vertical: 4),
          decoration: const BoxDecoration(color: Colors.black45),
          child: const Text(
            'Cuking',
            style: TextStyle(
              fontSize: 20,
              fontWeight: FontWeight.bold,
              color: Colors.white,
            ), // TextStyle
          ), // Text
        ), // Container
      ],
    ); // Stack
  }
}
```

zulfa

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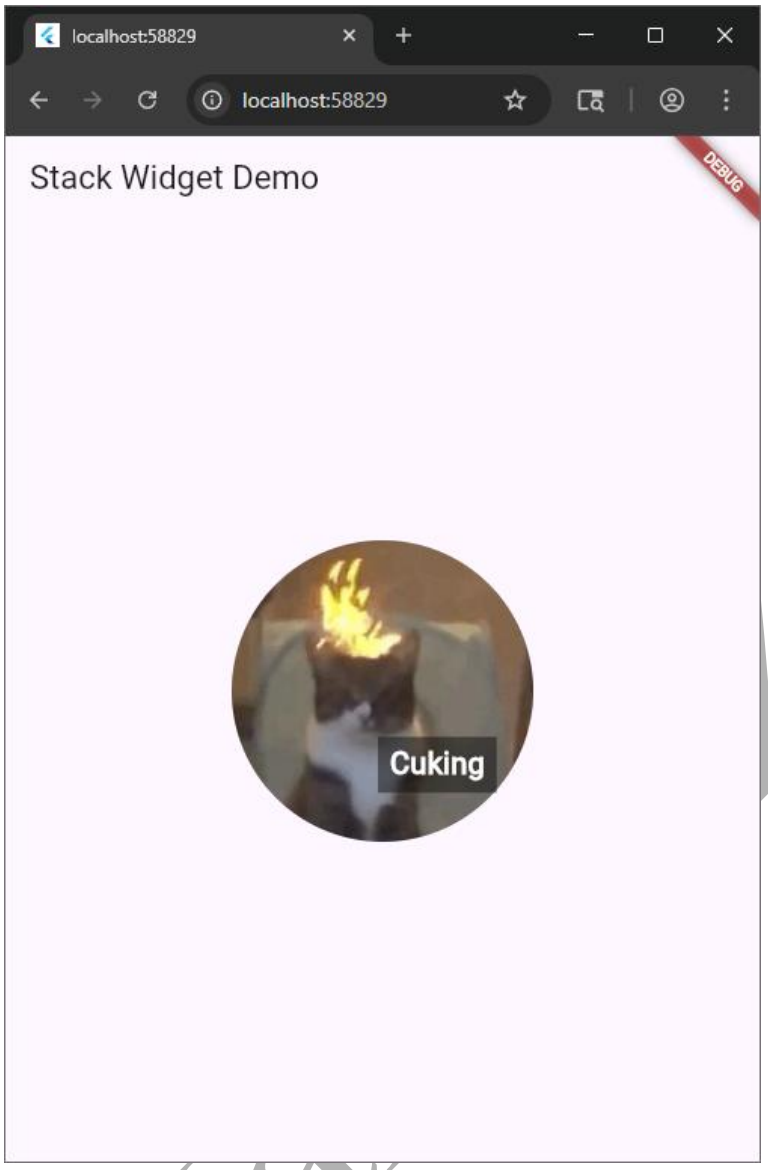
ABSENT : 11
CLASS : SIB 2G
OBJECT : Mobile Programming



Card

A Card, from the Material library, contains related nuggets of information and can be composed of almost any widget, but is often used with ListTile. Card has a single child, but its child can be a column, row, list, grid, or other widget that supports multiple children. By default, a Card shrinks its size to 0 by 0 pixels. You can use SizedBox to constrain the size of a card.

In Flutter, a Card features slightly rounded corners and a drop shadow, giving it a 3D effect. Changing a Card's elevation property allows you to control the drop shadow effect. Setting the elevation to 24, for example, visually lifts the Card further from the surface and causes the shadow to become more dispersed. For a list of supported elevation values, see Elevation in the Material guidelines. Specifying an unsupported value disables the drop shadow entirely.

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Summary (Card)

- Implements a Material card
- Used for presenting related nuggets of information
- Accepts a single child, but that child can be a Row, Column, or other widget that holds a list of children
- Displayed with rounded corners and a drop shadow
- A Card's content can't scroll
- From the Material library

card_demo.dart > CardDemo

```dart
import 'package:flutter/material.dart';

class CardDemo extends StatelessWidget {
  const CardDemo({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Card Widget Demo'),
      ), // AppBar
      backgroundColor: Colors.grey[200],
      body: Center(
        child: _buildCard(),
      ), // Center
    ); // Scaffold
  }
```

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OBJECT : Mobile Programming

```
Widget _buildCard() {
  return SizedBox(
    height: 210,
    child: Card(
      margin: const EdgeInsets.symmetric(horizontal: 20),
      elevation: 4,
      child: Column(
        children: [
          ListTile(
            title: const Text(
              '1625 Main Street',
              style: TextStyle(fontWeight: FontWeight.w500),
            ), // Text
            subtitle: const Text('My City, CA 99984'),
            leading: Icon(Icons.restaurant_menu, color: Colors.blue[500]),
          ), // ListTile
          const Divider(),
          ListTile(
            title: const Text(
              '(408) 555-1212',
              style: TextStyle(fontWeight: FontWeight.w500),
            ), // Text
            leading: Icon(Icons.contact_phone, color: Colors.blue[500]),
          ), // ListTile
          ListTile(
            title: const Text('costa@example.com'),
            leading: Icon(Icons.contact_mail, color: Colors.blue[500]),
          ), // ListTile
        ],
      ), // Column
    ), // Card
  ); // SizedBox
}
```

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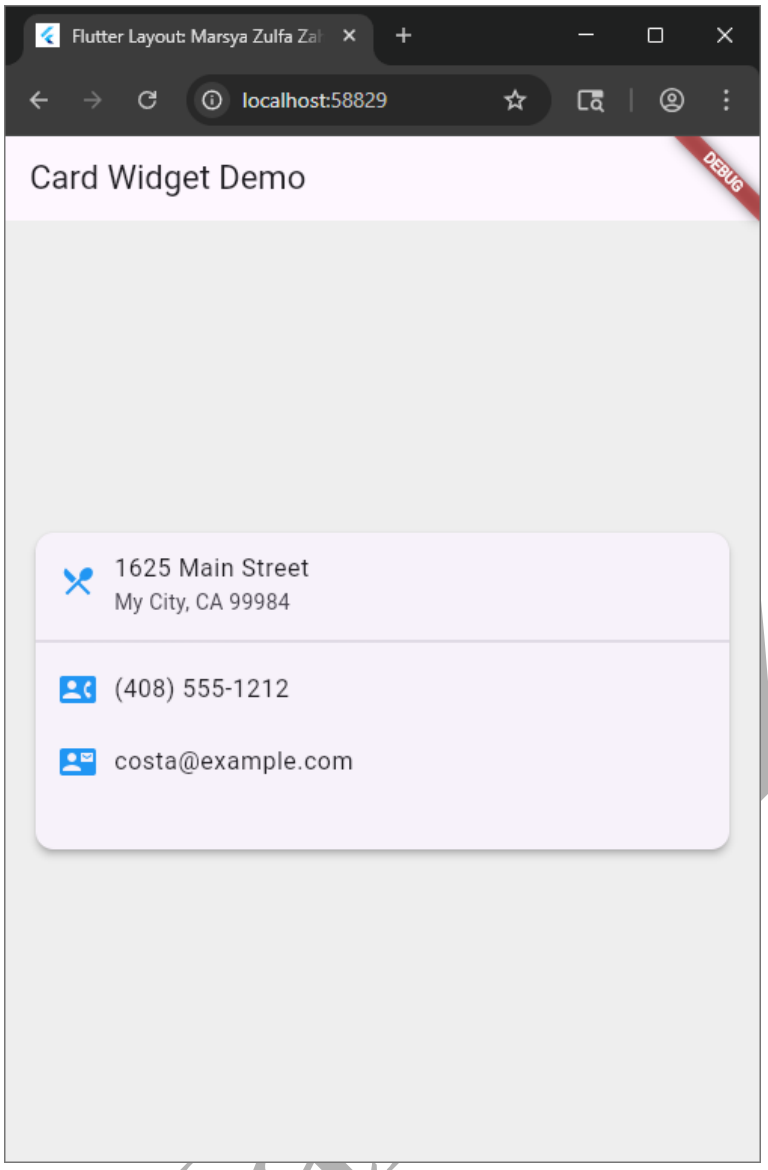

ListTile

Use ListTile, a specialized row widget from the Material library, for an easy way to create a row containing up to 3 lines of text and optional leading and trailing icons. ListTile is most commonly used in Card or ListView, but can be used elsewhere.

Summary (ListTile)

- A specialized row that contains up to 3 lines of text and optional icons
- Less configurable than Row, but easier to use
- From the Material library

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OBJECT : Mobile Programming

list_demo.dart > ListDemo

```dart
import 'package:flutter/material.dart';

class ListDemo extends StatelessWidget {
  const ListDemo({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('ListView dengan Nomor'),
      ), // AppBar
      body: _buildListView(),
    ); // Scaffold
  }
```

```dart
  Widget _buildListView() {
    return ListView.builder(
      padding: const EdgeInsets.all(8),
      itemCount: 4,
      itemBuilder: (context, index) {
        int itemNumber = index + 1;

        return ListTile(
          leading: CircleAvatar(
            backgroundColor: Colors.blue[100],
            child: Text(
              '$itemNumber',
              style: const TextStyle(color: Colors.black54),
            ), // Text
          ), // CircleAvatar
          title: Text(
            'Item $itemNumber',
            style: const TextStyle(fontWeight: FontWeight.w500),
          ), // Text
          subtitle: const Text('Secondary text'),
        ); // ListTile
      },
    ); // ListView.builder
  }
}
```

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CLASS : SIB 2G
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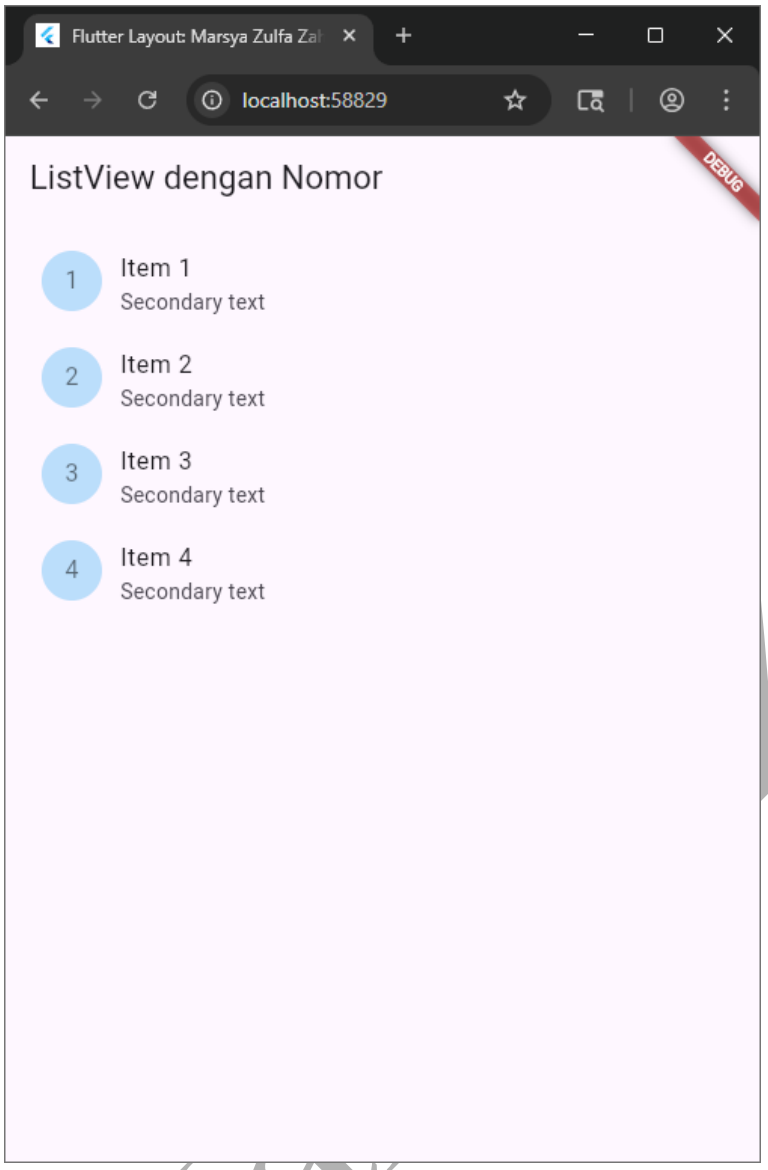
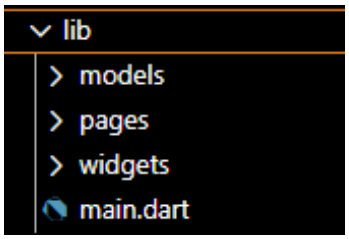


Praktikum 5: Membangun Navigasi di Flutter

Langkah 1: Siapkan project baru

Sebelum melanjutkan praktikum, buatlah sebuah project baru Flutter dengan nama belanja dan susunan folder seperti pada gambar berikut. Penyusunan ini dimaksudkan untuk mengorganisasi kode dan widget yang lebih mudah.



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Langkah 2: Mendefinisikan Route

Buatlah dua buah file dart dengan nama home_page.dart dan item_page.dart pada folder pages. Untuk masing-masing file, deklarasikan class HomePage pada file home_page.dart dan ItemPage pada item_page.dart. Turunkan class dari StatelessWidget. Gambaran potongan kode dapat anda lihat sebagai berikut.

```
pages > home_page.dart > ...
import 'package:flutter/material.dart';

class HomePage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    // TODO: implement build    TODO: imp
    throw UnimplementedError();
  }
}
```

```
pages > item_page.dart > ...
import 'package:flutter/material.dart';

class ItemPage extends StatelessWidget {    Construct
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Item Page')),
      body: Center(child: Text('Ini halaman Item')),
    ); // Scaffold
  }
}
```

Langkah 3: Lengkapi Kode di main.dart

Setelah kedua halaman telah dibuat dan didefinisikan, bukalah file main.dart. Pada langkah ini anda akan mendefinisikan Route untuk kedua halaman tersebut. Definisi penamaan route harus bersifat unique. Halaman HomePage didefinisikan sebagai /. Dan halaman ItemPage didefinisikan sebagai /item. Untuk mendefinisikan halaman awal, anda dapat menggunakan named argument initialRoute. Gambaran tahapan ini, dapat anda lihat pada potongan kode berikut.

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```
main.dart > ...
import 'package:flutter/material.dart';
import 'package:belanja/pages/home_page.dart';
import 'package:belanja/pages/item_page.dart';

Run | Debug | Profile
void main() {
  runApp(MaterialApp(
    initialRoute: '/',
    routes: {
      '/': (context) => HomePage(),
      '/item': (context) => ItemPage(),
    },
  )); // MaterialApp
}
```

Langkah 4: Membuat data model

Sebelum melakukan perpindahan halaman dari HomePage ke ItemPage, dibutuhkan proses pemodelan data. Pada desain mockup, dibutuhkan dua informasi yaitu nama dan harga. Untuk menangani hal ini, buatlah sebuah file dengan nama item.dart dan letakkan pada folder models. Pada file ini didefinisikan pemodelan data yang dibutuhkan. Ilustrasi kode yang dibutuhkan, dapat anda lihat pada potongan kode berikut.

```
models > item.dart > ...
class Item {
  String name;
  int price;

  Item({required this.name, required this.price});
}
```

Langkah 5: Lengkapi kode di class HomePage

Pada halaman HomePage terdapat ListView widget. Sumber data ListView diambil dari model List dari object Item. Gambaran kode yang dibutuhkan untuk melakukan definisi model dapat anda lihat sebagai berikut.

```
pages > home_page.dart > HomePage
import 'package:flutter/material.dart';
import 'package:belanja/models/item.dart';

class HomePage extends StatelessWidget {
  final List<Item> items = [
    Item(name: 'Sugar', price: 5000),
    Item(name: 'Salt', price: 2000)
  ];
```

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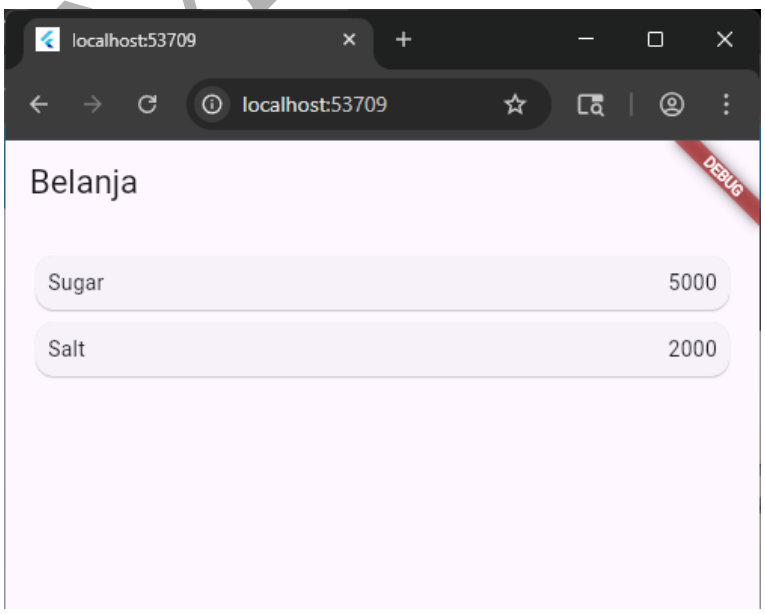
Langkah 6: Membuat ListView dan itemBuilder

Untuk menampilkan ListView pada praktikum ini digunakan itemBuilder. Data diambil dari definisi model yang telah dibuat sebelumnya. Untuk menunjukkan batas data satu dan berikutnya digunakan widget Card. Kode yang telah umum pada bagian ini tidak ditampilkan. Gambaran kode yang dibutuhkan dapat anda lihat sebagai berikut.

```
body: Container(
  margin: EdgeInsets.all(8),
  child: ListView.builder(
    padding: EdgeInsets.all(8),
    itemCount: items.length,
    itemBuilder: (context, index) {
      final item = items[index];
        return Card(
          child: Container(
            margin: EdgeInsets.all(8),
            child: Row(
              children: [
                Expanded(child: Text(item.name)),
                Expanded(
                  child: Text(
                    item.price.toString(),
                    textAlign: TextAlign.end,
                  ), // Text
                ), // Expanded
              ],
            ), // Row
          ), // Container
        ); // Card
    },
  ), // ListView.builder
), // Container
```

Jalankan aplikasi pada emulator atau pada device anda.



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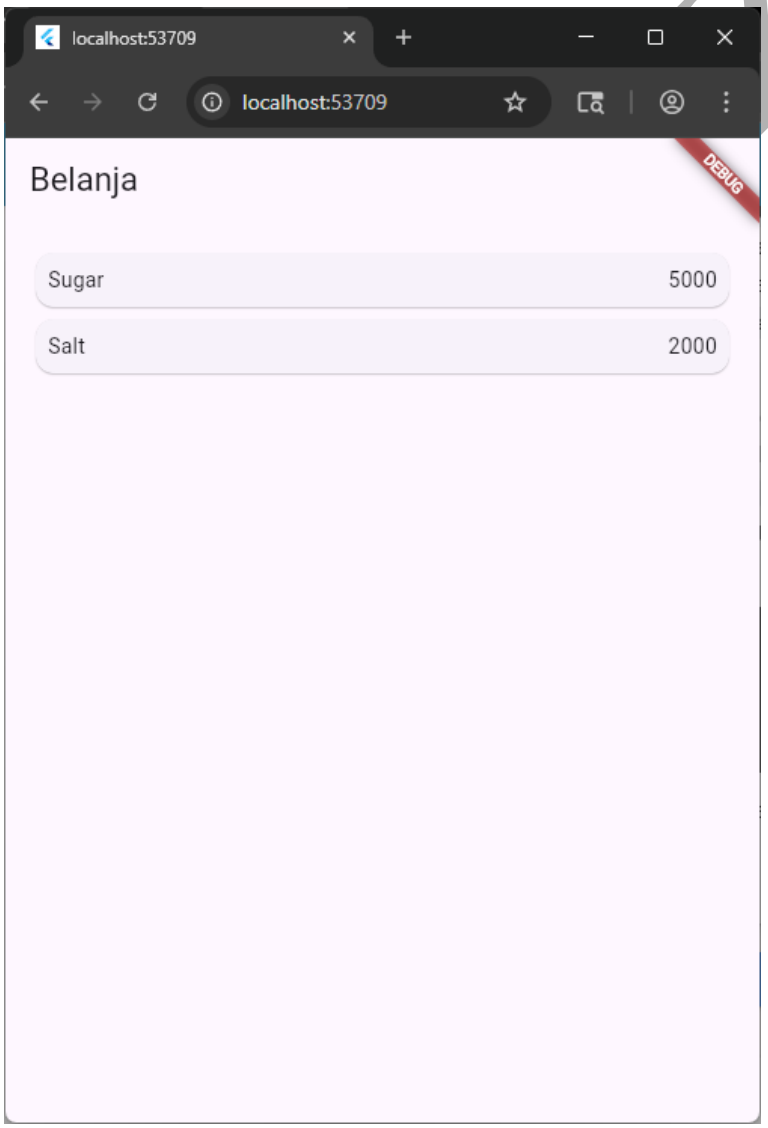
Langkah 7: Menambahkan aksi pada ListView

Item pada ListView saat ini ketika ditekan masih belum memberikan aksi tertentu. Untuk menambahkan aksi pada ListView dapat digunakan widget InkWell atau GestureDetector. Perbedaan utamanya InkWell merupakan material widget yang memberikan efek ketika ditekan. Sedangkan GestureDetector bersifat umum dan bisa juga digunakan untuk gesture lain selain sentuhan. Pada praktikum ini akan digunakan widget InkWell.

Untuk menambahkan sentuhan, letakkan cursor pada widget pembuka Card. Kemudian gunakan shortcut quick fix dari VSCode (Ctrl + . pada Windows atau Cmd + . pada MacOS). Sorot menu wrap with widget... Ubah nilai widget menjadi InkWell serta tambahkan named argument onTap yang berisi fungsi untuk berpindah ke halaman ItemPage. Ilustrasi potongan kode dapat anda lihat pada potongan berikut.

```
return InkWell(
  onTap: () {
    Navigator.pushNamed(context, '/item');
  },
```



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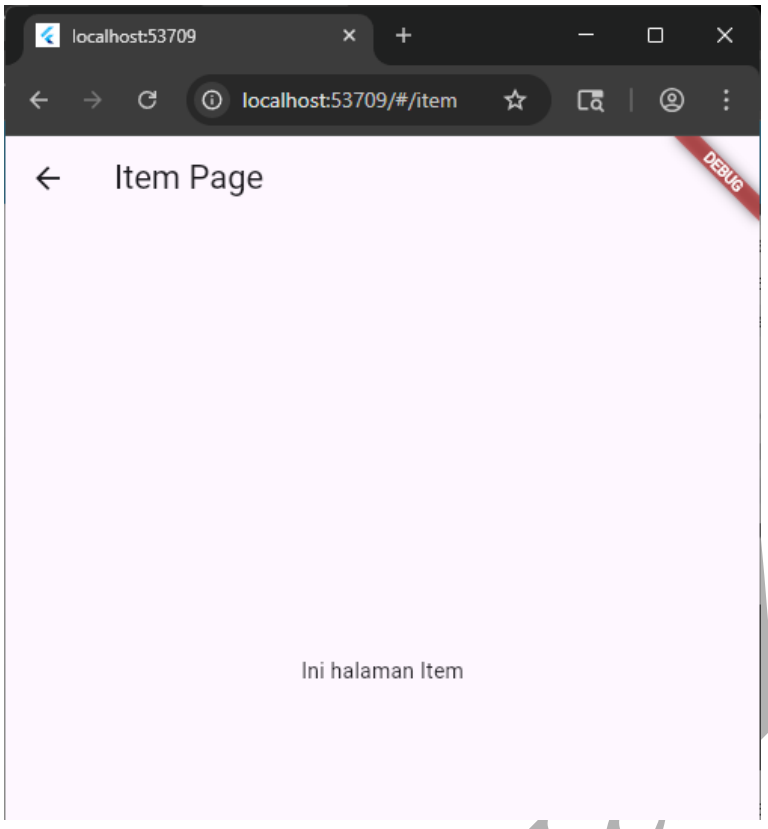
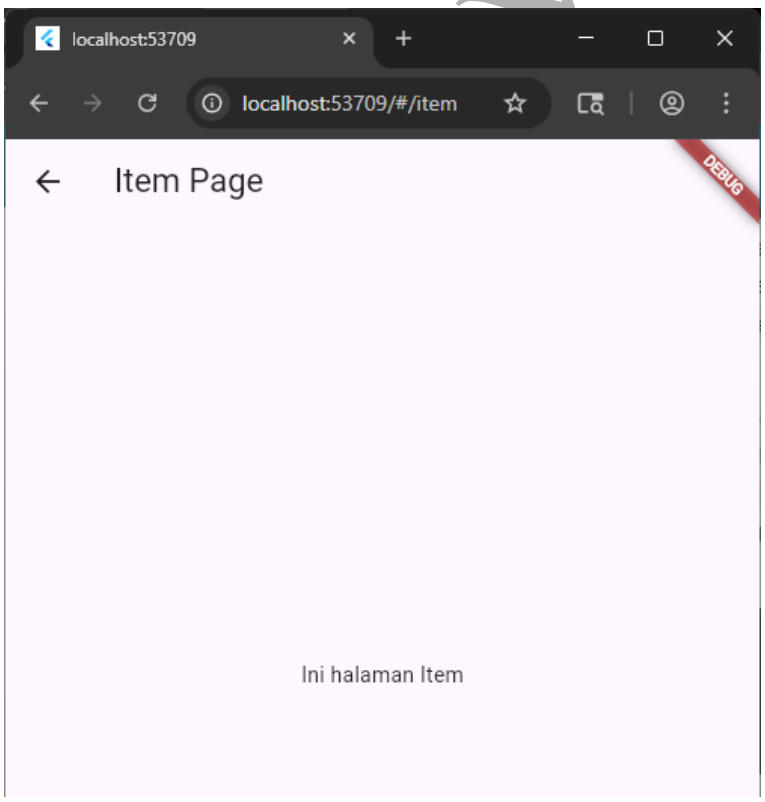
Klik on Sugar



Klik on Salt



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CLASS : SIB 2G
OBJECT : Mobile Programming

Tugas Praktikum 2

1. Untuk melakukan pengiriman data ke halaman berikutnya, cukup menambahkan informasi arguments pada penggunaan Navigator. Perbarui kode pada bagian Navigator menjadi seperti berikut.

```
return InkWell(
  onTap: () {
    Navigator.pushNamed(context, '/item', arguments: item);
  },
```

2. Pembacaan nilai yang dikirimkan pada halaman sebelumnya dapat dilakukan menggunakan ModalRoute. Tambahkan kode berikut pada blok fungsi build dalam halaman ItemPage. Setelah nilai didapatkan, anda dapat menggunakannya seperti penggunaan variabel pada umumnya. (https://docs.flutter.dev/cookbook/navigation/navigate-with-arguments)

```
final itemArgs = ModalRoute.of(context)!.settings.arguments as Item;
```

3. Pada hasil akhir dari aplikasi belanja yang telah anda selesaikan, tambahkan atribut foto produk, stok, dan rating. Ubahlah tampilan menjadi GridView seperti di aplikasi marketplace pada umumnya.

```
models > item.dart > ...
class Item {
  String name;
  int price;
  String imageUrl;
  int stock;
  double rating;

  Item({
    required this.name,
    required this.price,
    required this.imageUrl,
    required this.stock,
    required this.rating,
  });
}
```

Marsya Zulfa

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CLASS : SIB 2G
OBJECT : Mobile Programming

```
body: Container(
  margin: const EdgeInsets.all(8),
  child: GridView.builder(
    gridDelegate: const SliverGridDelegateWithFixedCrossAxisCount(
      crossAxisCount: 2,
      childAspectRatio: 0.7,
      crossAxisSpacing: 10,
      mainAxisSpacing: 10,
    ), // SliverGridDelegateWithFixedCrossAxisCount
    itemCount: items.length,
    itemBuilder: (context, index) {
      final item = items[index];
      return ItemCard(item: item);
    },
  ), // GridView.builder
), // Container
```

4. Silakan implementasikan Hero widget pada aplikasi belanja Anda dengan mempelajari dari sumber ini: https://docs.flutter.dev/cookbook/navigation/hero-animations

```
Hero(
  tag: item.name,
  child: ClipRRect(
    borderRadius: const BorderRadius.vertical(top: Radius.circular(15)),
    child: Image.network(
      item.imageUrl,
      height: 120,
      width: double.infinity,
      fit: BoxFit.cover,
    ), // Image.network
  ), // ClipRRect
), // Hero
```

```
Hero(
  tag: itemArgs.name,
  child: Image.network(
    itemArgs.imageUrl,
    width: double.infinity,
    height: 300,
    fit: BoxFit.cover,
  ), // Image.network
), // Hero
```

5. Sesuaikan dan modifikasi tampilan sehingga menjadi aplikasi yang menarik. Selain itu, pecah widget menjadi kode yang lebih kecil. Tambahkan Nama dan NIM di footer aplikasi belanja Anda.
Merapikan widget

```
return ItemCard(item: item);
```

```
class ItemCard extends StatelessWidget {
  final Item item;

  const ItemCard({Key? key, required this.item}) : super(key: key);
```

NAME : MARSYA ZULFA ZAHRANI
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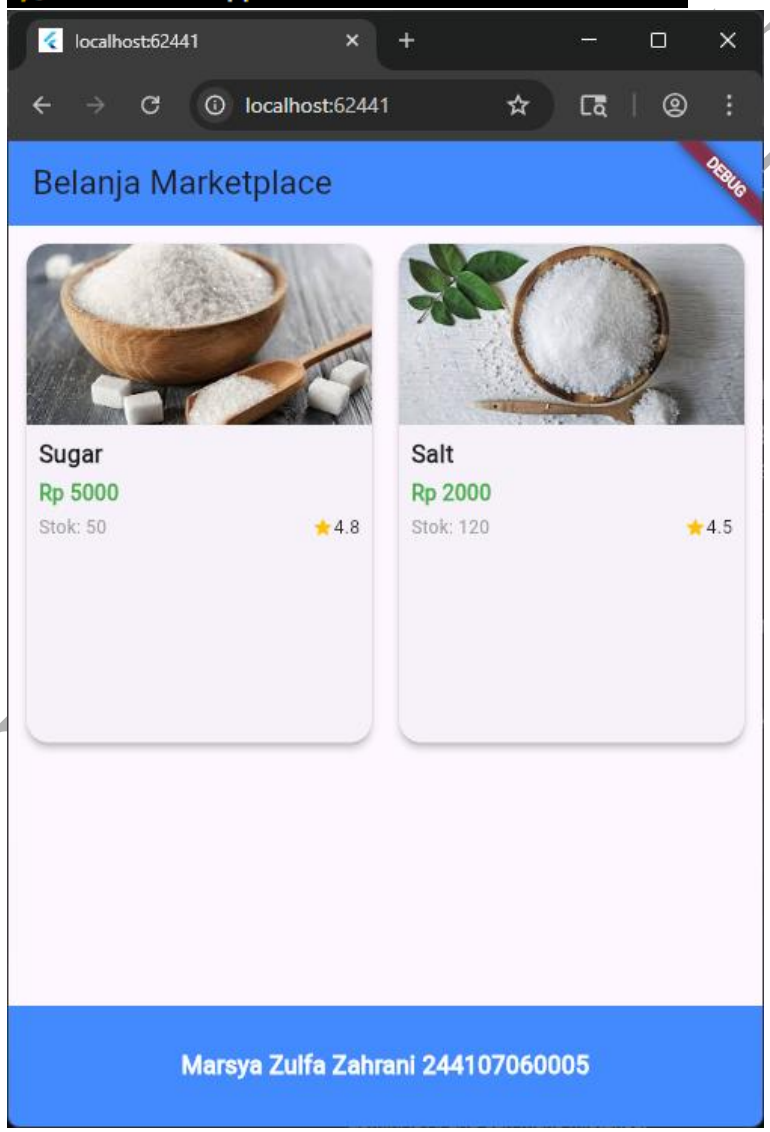
ABSENT : 11
CLASS : SIB 2G
OBJECT : Mobile Programming

```
bottomNavigationBar: const BottomAppBar(
  color: Colors.blueAccent,
  child: Padding(
    padding: EdgeInsets.all(16.0),
    child: Text(
      'Marsya Zulfa Zahrani 244107060005',
      textAlign: TextAlign.center,
      style: TextStyle(
        color: Colors.white,
        fontWeight: FontWeight.bold,
        fontSize: 16,
      ), // TextStyle
    ), // Text
  ), // Padding
), // BottomAppBar
```



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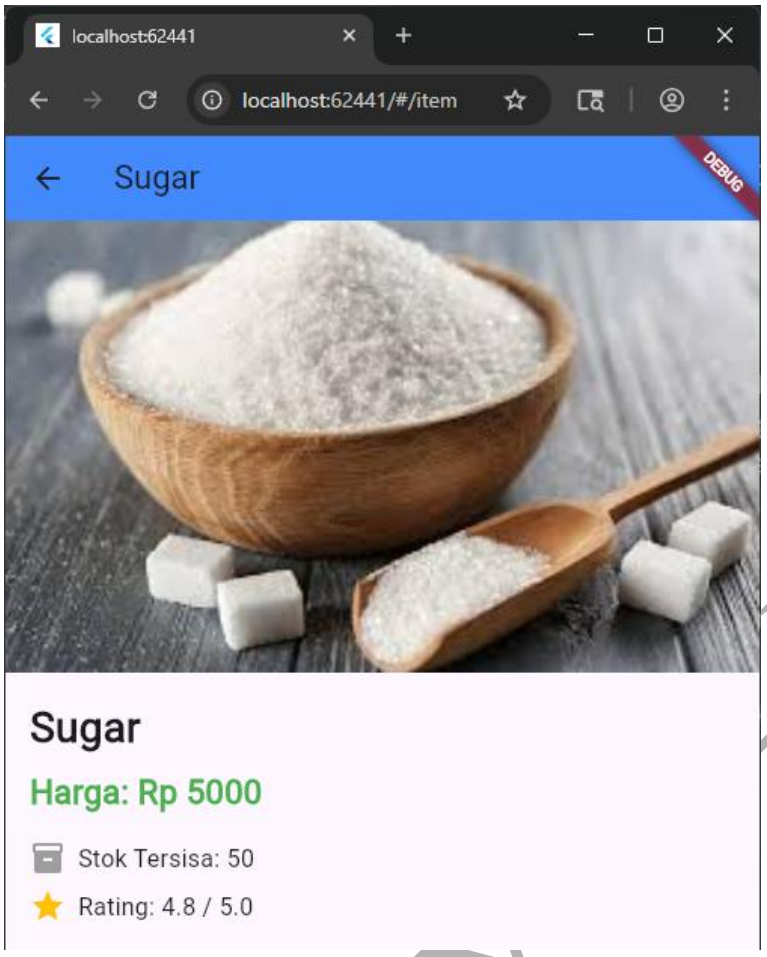
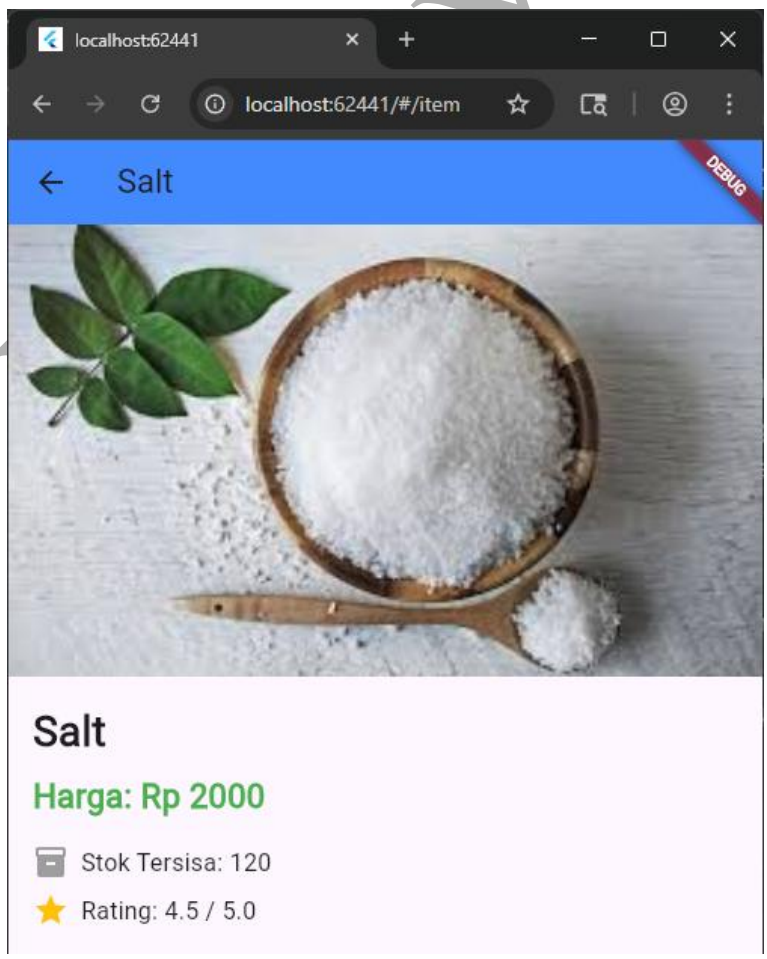
CLASS : SIB 2G
OBJECT : Mobile Programming





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6. Selesaikan Praktikum 5: Navigasi dan Rute tersebut. Cobalah modifikasi menggunakan plugin go_router, lalu dokumentasikan dan push ke repository Anda berupa screenshot setiap hasil pekerjaan beserta penjelasannya di file README.md. Kumpulkan link commit repository GitHub Anda kepada dosen yang telah disepakati!

main.dart > MyApp

```dart
import 'package:flutter/material.dart';
import 'package:go_router/go_router.dart';
import 'package:belanja/pages/home_page.dart';
import 'package:belanja/pages/item_page.dart';
import 'package:belanja/models/item.dart';
Run | Debug | Profile
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);
  @override
  Widget build(BuildContext context) {
    final GoRouter router = GoRouter(
      initialLocation: '/',
      routes: [
        GoRoute(
          path: '/',
          builder: (context, state) => HomePage(),
        ), // GoRoute
        GoRoute(
          path: '/item',
          builder: (context, state) {
            final Item itemArgs = state.extra as Item;
            return ItemPage(itemArgs: itemArgs);
          },
        ), // GoRoute
      ],
    ); // GoRouter
    return MaterialApp.router(
      title: 'Belanja App',
      routerConfig: router,
    );
  }
```

pages > item_page.dart > ItemPage

```dart
import 'package:flutter/material.dart';
import 'package:belanja/models/item.dart';

class ItemPage extends StatelessWidget {
  final Item itemArgs;

  const ItemPage({Key? key, required this.itemArgs}) : super(key: key);
```

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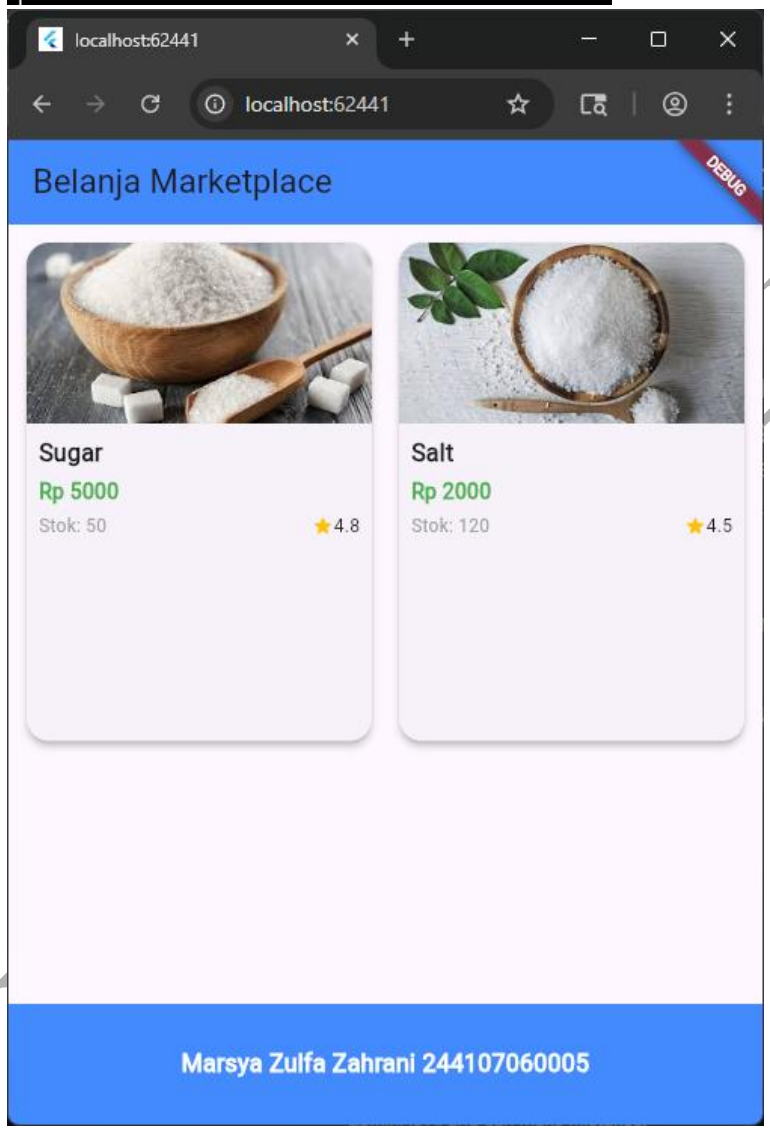
pages > home_page.dart > HomePage

```dart
import 'package:flutter/material.dart';
import 'package:go_router/go_router.dart';
import 'package:belanja/models/item.dart';
```

```dart
return InkWell(
  onTap: () {
    context.go('/item', extra: item);
  },
```



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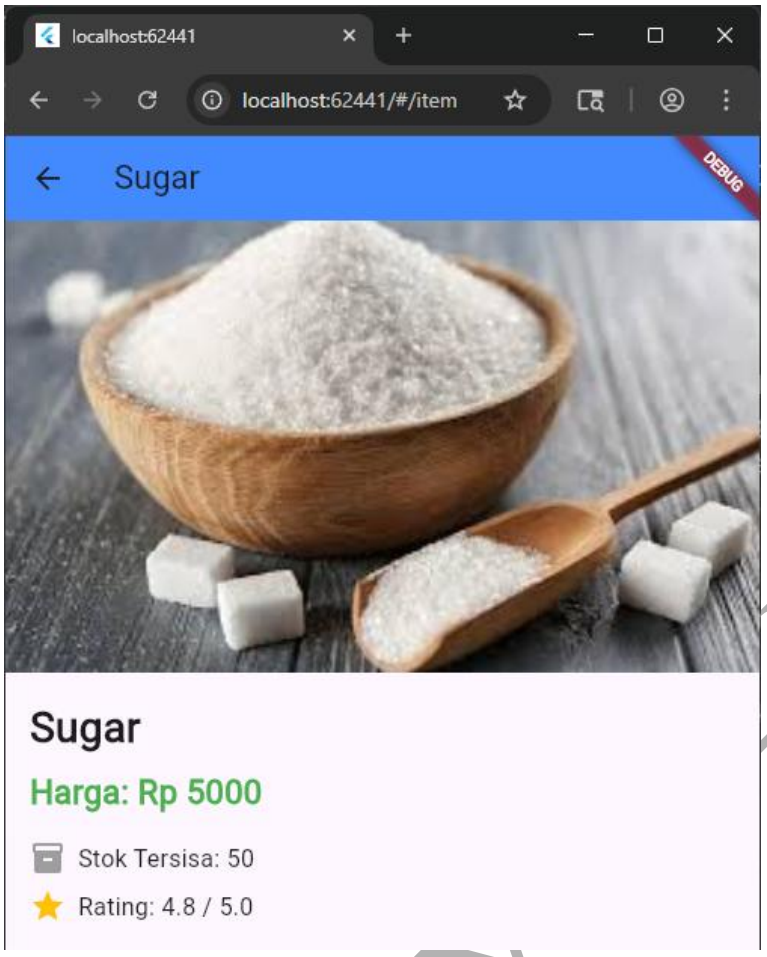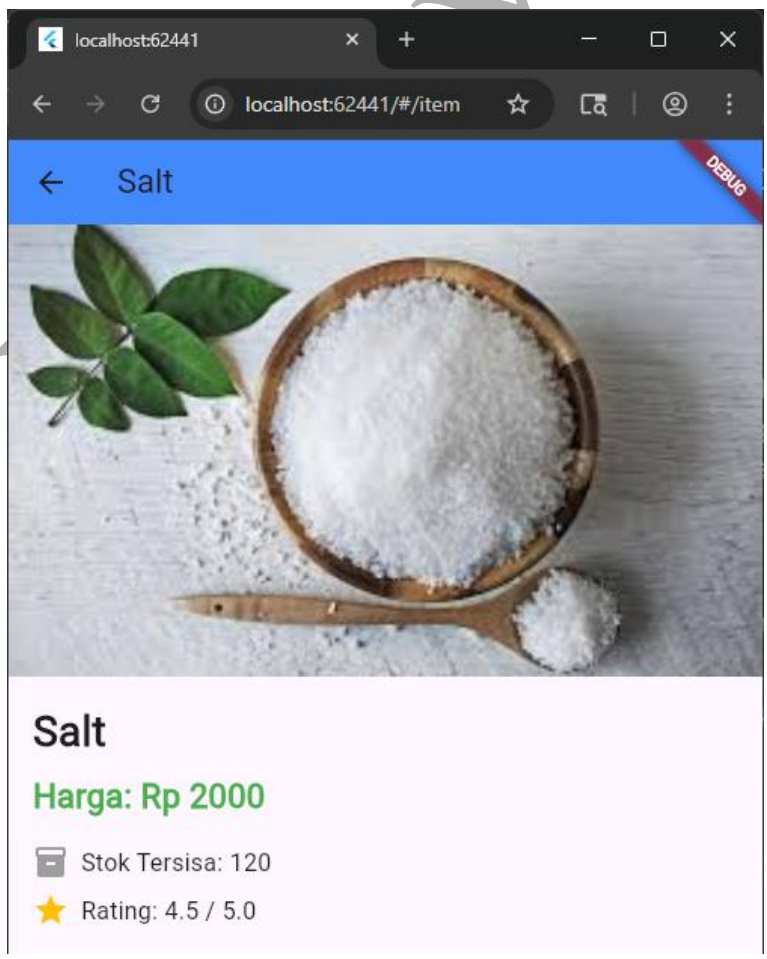
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